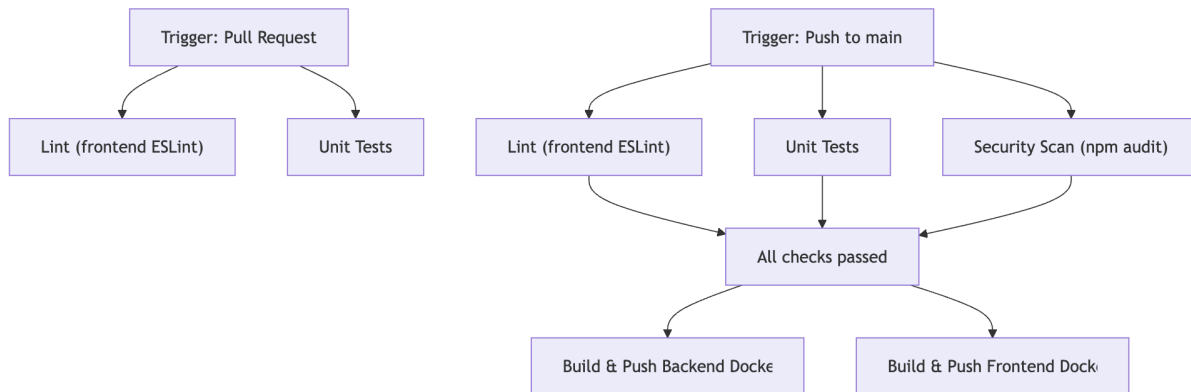


Continuous Integration

Diagram



CI Pipeline Description

This CI pipeline has two trigger paths. When a pull request is opened or updated, the pipeline runs frontend linting and unit tests in parallel. This helps catch code quality issues and functional regressions before changes are merged.

When code is pushed to the main branch, the pipeline runs frontend linting, unit tests, and a dependency security scan in parallel. If all checks pass, the pipeline proceeds to build and push both the backend and frontend Docker images to Docker Hub using the latest tag. This design improves efficiency by allowing independent checks to run concurrently while ensuring that container images are only published after all required validations succeed.

Code Quality Checks

description

We configured the CI pipeline to automatically run a linter whenever code is pushed or a pull request is opened. In our project, this is implemented using ESLint for the frontend service. The workflow installs the frontend dependencies and runs `npm run lint` in the frontend directory as part of the GitHub Actions pipeline.

This check helps detect code quality issues such as unused variables, invalid patterns, and violations of the configured ESLint rules before code is merged into the main codebase. If the lint step fails, the CI pipeline reports the failure immediately, ensuring that code quality problems are caught early in the development process.

← CI — Lint & unit tests

✖ Introduce lint failure for CI proof #8

Re-run jobs

Summary

All jobs

✖ ESLint (frontend)

- Unit tests
- Security scan (npm audit)
- Build & push backend image
- Build & push frontend image

Run details

Usage

Workflow file

Annotations

2 errors

ESLint (frontend)

failed 4 minutes ago in 19s

Search logs

Explain error



>	Set up job	1s
>	Run actions/checkout@v6	1s
>	Run actions/setup-node@v6	3s
>	Install frontend dependencies	10s
▼	Run ESLint	2s
1	▶ Run npm run lint	
4		
5	> frontend@0.0.0 lint	
6	> eslint . --ext js,jsx --report-unused-disable-directives --max-warnings 0	
7		
8		
9	/home/runner/work/HealthEduLtd/HealthEduLtd/frontend/src/App.jsx	
10	Error: 10:9 error 'lintFailureDemo' is assigned a value but never used. Allowed unused vars must match /^React\$/u no-unused-vars	
11		
12	✖ 1 problem (1 error, 0 warnings)	
13		
14	Error: Process completed with exit code 1.	
	Post Run actions/setup-node@v6	0s

Unit testing

description

We configured the CI pipeline to automatically run unit tests whenever a pull request is opened or when code is pushed to the main branch. The pipeline executes both backend and frontend tests through the npm test command in GitHub Actions.

This ensures that functional regressions are detected early before code is merged or released from the main branch. If any unit test fails, the Unit tests job fails, which causes the CI pipeline to fail as well. This behavior enforces test correctness and prevents broken code from passing the CI process.

✖ Introduce unit test failure for CI proof #10

Re-run jobs ⌵ ⋮

Summary

All jobs

✔ ESLint (frontend)

✖ Unit tests

🕒 Security scan (npm audit)

🕒 Build & push backend image

🕒 Build & push frontend image

Run details

🕒 Usage

📄 Workflow file

Annotations

2 errors

Unit tests

failed 4 minutes ago in 41s

🔍 Search logs

📄 Explain error

🔄

⚙️

> ✔ Set up job

1s

> ✔ Run actions/checkout@v6

1s

> ✔ Run actions/setup-node@v6

4s

> ✔ Install root dependencies

23s

> ✔ Install frontend dependencies

7s

✖ Run unit tests

1s

Summary

All jobs

✔ ESLint (frontend)

✖ Unit tests

🕒 Security scan (npm audit)

🕒 Build & push backend image

🕒 Build & push frontend image

Run details

🕒 Usage

📄 Workflow file

Unit tests

failed 4 minutes ago in 41s

🔍 Search logs

📄 Explain error

🔄

⚙️

✖ Run unit tests

1s

```
122 ----- Failed Tests 1 -----
123
124 FAIL backend/utils/paymentEnv.test.js > paymentEnv > getFrontendBaseUrl > defaults to Vite dev URL
125 AssertionError: expected 'http://localhost:5173' to be 'http://localhost:9999' // Object.is equality
126
127 Expected: "http://localhost:9999"
128 Received: "http://localhost:5173"
129
130 > backend/utils/paymentEnv.test.js:20:36
131   18|   describe("getFrontendBaseUrl", () => {
132   19|     it("defaults to Vite dev URL", () => {
133   20|       expect(getFrontendBaseUrl()).toBe("http://localhost:9999");
134     |                                     ^
135   21|     });
136   22|   });
137
138 -----[1/1]-----
139
140
141 Test Files  1 failed | 6 passed (7)
142 Tests      1 failed | 39 passed (40)
143 Start at   20:19:00
144 Duration  1.02s (transform 333ms, setup 0ms, import 461ms, tests 775ms, environment 1ms)
145
146
147 Error: AssertionError: expected 'http://localhost:5173' to be 'http://localhost:9999' // Object.is equality
148
149 Expected: "http://localhost:9999"
150 Received: "http://localhost:5173"
151
152 > backend/utils/paymentEnv.test.js:20:36
153
154
155 Error: Process completed with exit code 1.
```

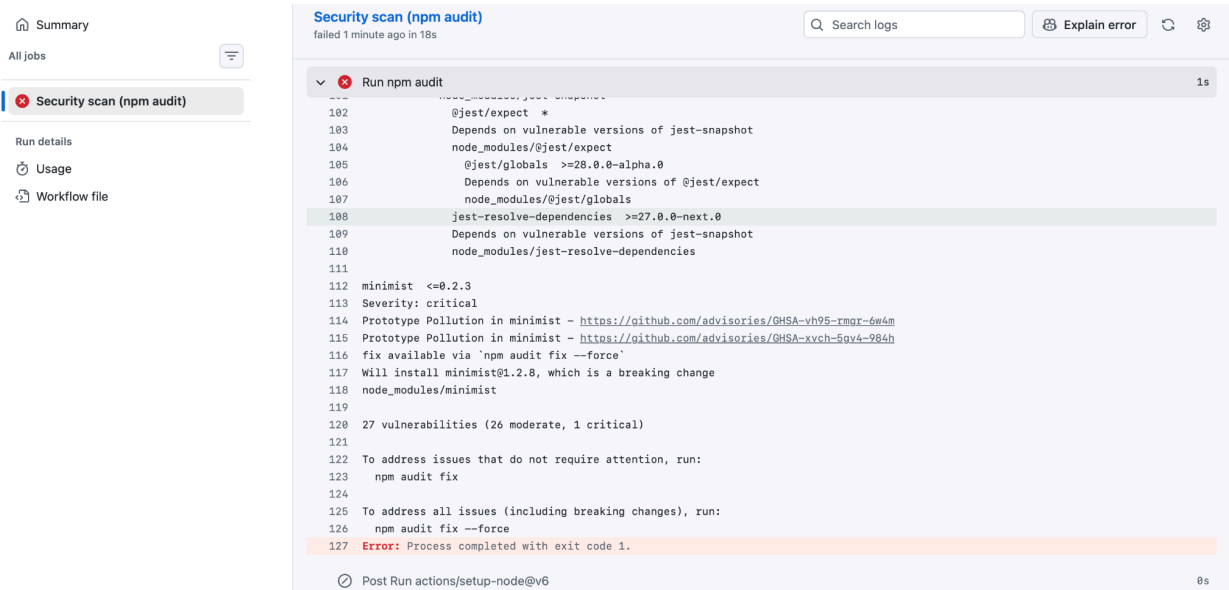
Security Scan

Description

We configured the CI pipeline to automatically run a dependency vulnerability scan whenever code is pushed to the main branch. In our project, this is implemented using npm audit on the root Node.js service, which represents the backend service dependencies.

This security scan helps identify known high-severity vulnerabilities in project dependencies before deployment. If npm audit detects a vulnerability at or above the configured threshold, the

Security scan job fails and the CI pipeline reports the failure. This ensures that dependency-related security issues are caught before container images are built and pushed.



The screenshot displays a GitHub Actions workflow run for a job named "Security scan (npm audit)". The job status is "failed 1 minute ago in 18s". The left sidebar shows the job's summary, details, usage, and workflow file. The main panel shows the output of the "Run npm audit" step, which failed with exit code 1. The output text includes:

```
102 @jest/expect *
103 Depends on vulnerable versions of jest-snapshot
104 node_modules/@jest/expect
105 @jest/globals >=28.0.0-alpha.0
106 Depends on vulnerable versions of @jest/expect
107 node_modules/@jest/globals
108 jest-resolve-dependencies >=27.0.0-next.0
109 Depends on vulnerable versions of jest-snapshot
110 node_modules/jest-resolve-dependencies
111
112 minimist <=0.2.3
113 Severity: critical
114 Prototype Pollution in minimist - https://github.com/advisories/GHSA-vh95-rmqr-6w4m
115 Prototype Pollution in minimist - https://github.com/advisories/GHSA-xvch-5gv4-984h
116 fix available via `npm audit fix --force`
117 Will install minimist@1.2.8, which is a breaking change
118 node_modules/minimist
119
120 27 vulnerabilities (26 moderate, 1 critical)
121
122 To address issues that do not require attention, run:
123   npm audit fix
124
125 To address all issues (including breaking changes), run:
126   npm audit fix --force
127 Error: Process completed with exit code 1.
```

Below the output, it shows the step "Post Run actions/setup-node@v6" with a duration of 0s.

CI Pipeline Code Links

The CI pipeline configuration is implemented in the GitHub Actions workflow file below, and the container build definitions are provided in the backend and frontend Dockerfiles.

- CI workflow:
<https://github.com/yuananbo/HealthEduLtd/blob/feature/ci-lint-and-tests/.github/workflows/ci-lint-and-test.yml>
- Backend Dockerfile:
<https://github.com/yuananbo/HealthEduLtd/blob/feature/ci-lint-and-tests/Dockerfile>
- Frontend Dockerfile:
<https://github.com/yuananbo/HealthEduLtd/blob/feature/ci-lint-and-tests/frontend/Dockerfile>